CIS 5450 Final Project Presentation

Employment Outcome Prediction

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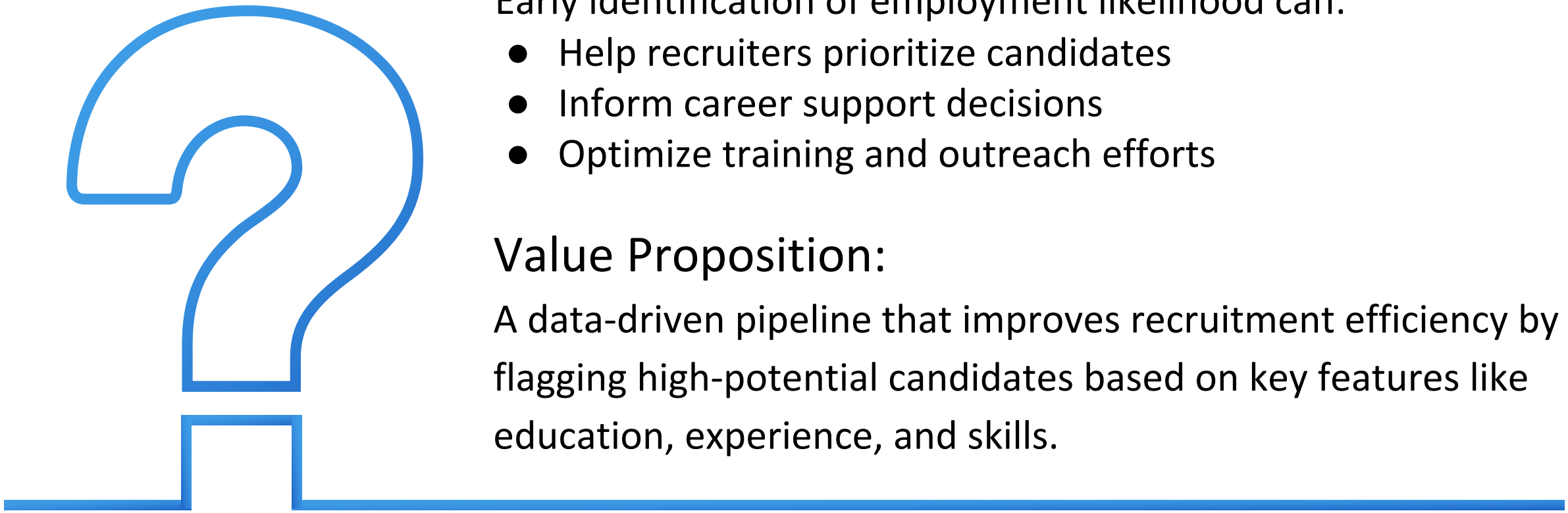
Objective & Value Proposition

Objective:

Predict whether a job applicant is employed based on their profile information.



Why it matters:

Early identification of employment likelihood can:

- Help recruiters prioritize candidates
- Inform career support decisions
- Optimize training and outreach efforts

Value Proposition:

A data-driven pipeline that improves recruitment efficiency by flagging high-potential candidates based on key features like education, experience, and skills.

Dataset Overview

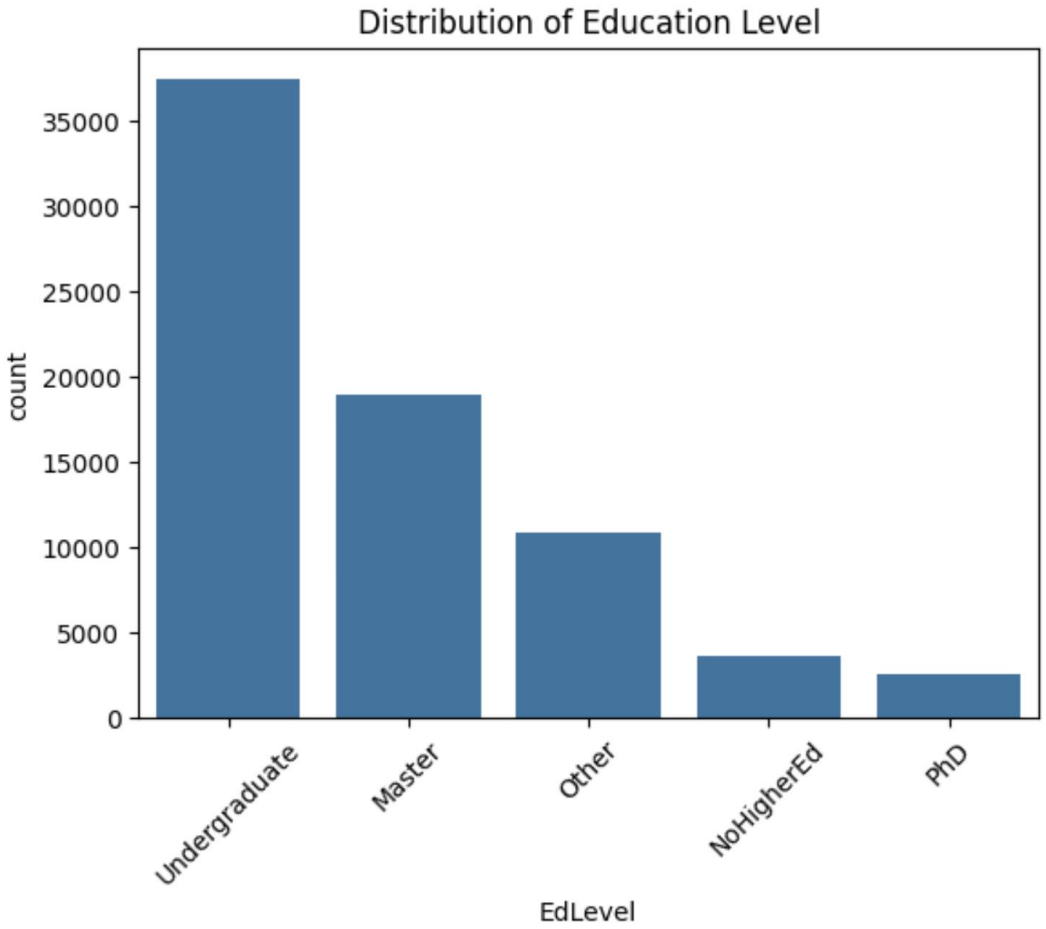
Source:
public dataset from Kaggle (https://www.kaggle.com/datasets/ayushtankha/70k-job-applicants-data-human-resource)

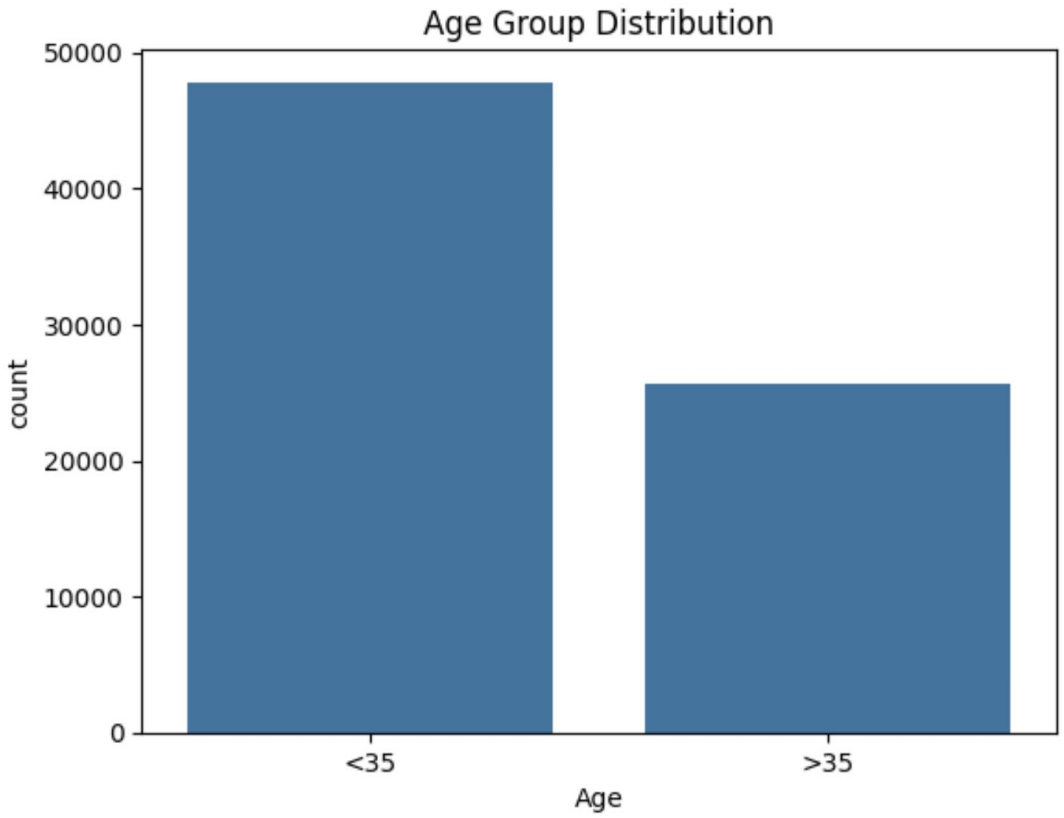
Size & Format:
~70,000 rows,
15 features,
structured tabular format

Age	Accessibility	EdLevel	Employment	Gender	MentalHealth	MainBranch	YearsCode	YearsCodePro	Country	PreviousSalary

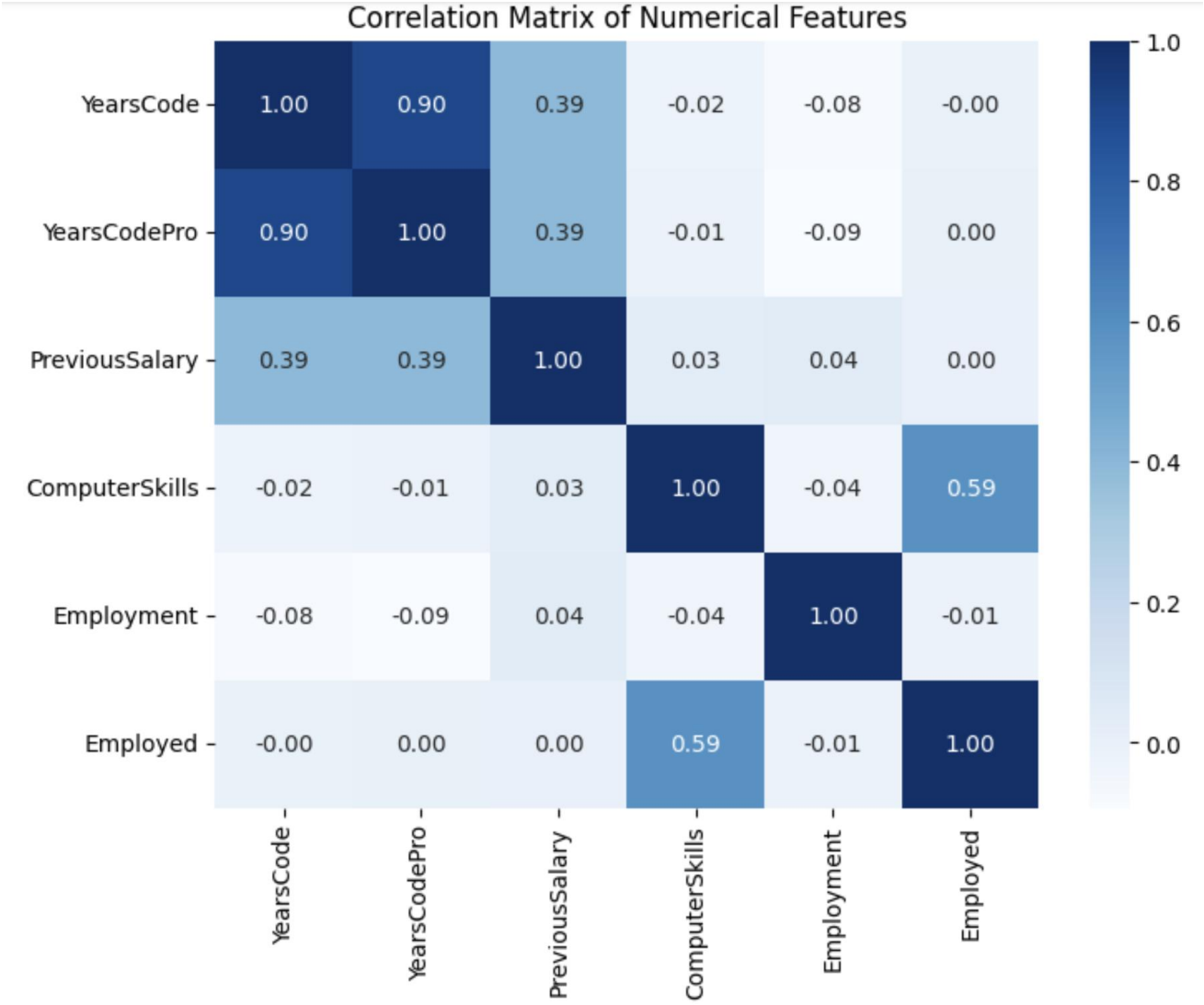
Target variable:
Employed (0 or 1)

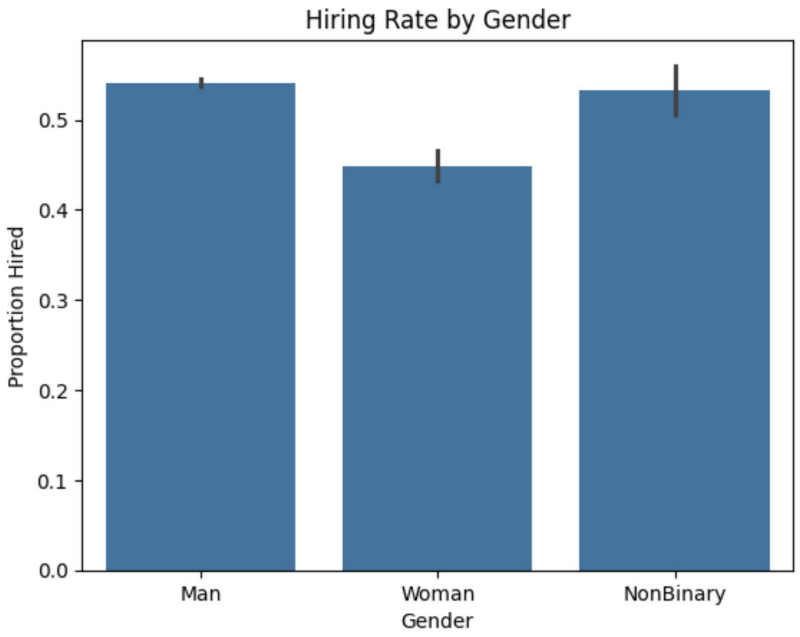
Feature Distributions

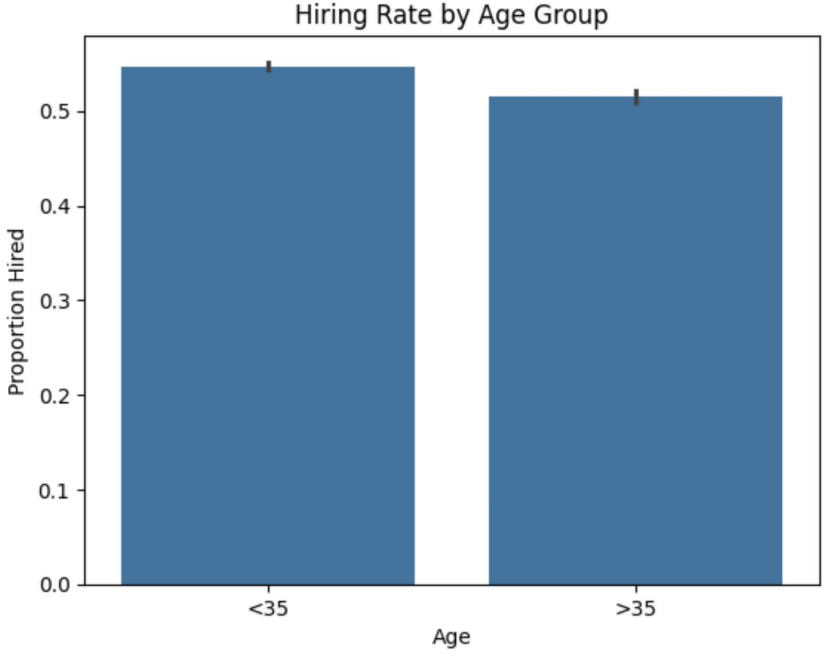




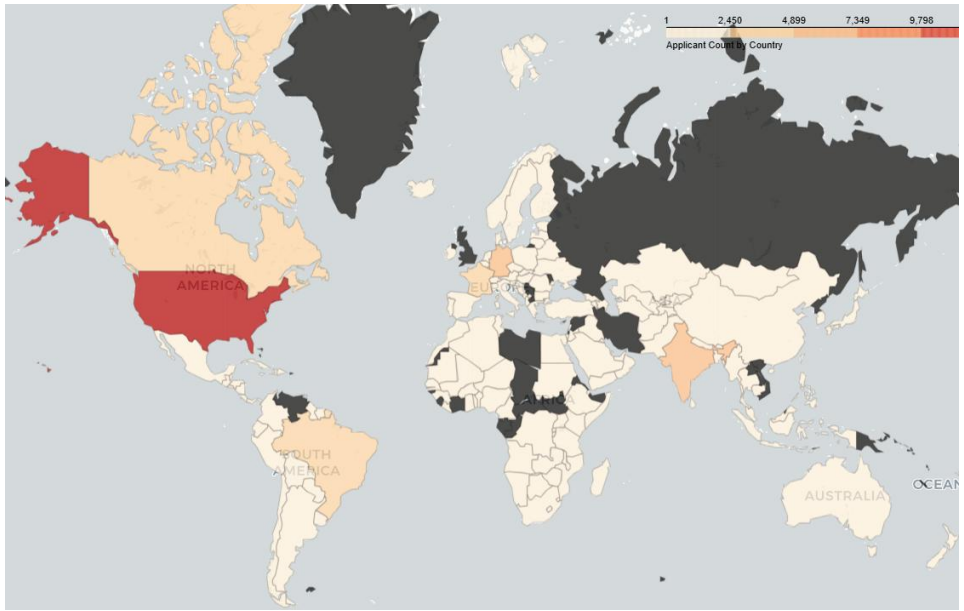
Correlation Analysis







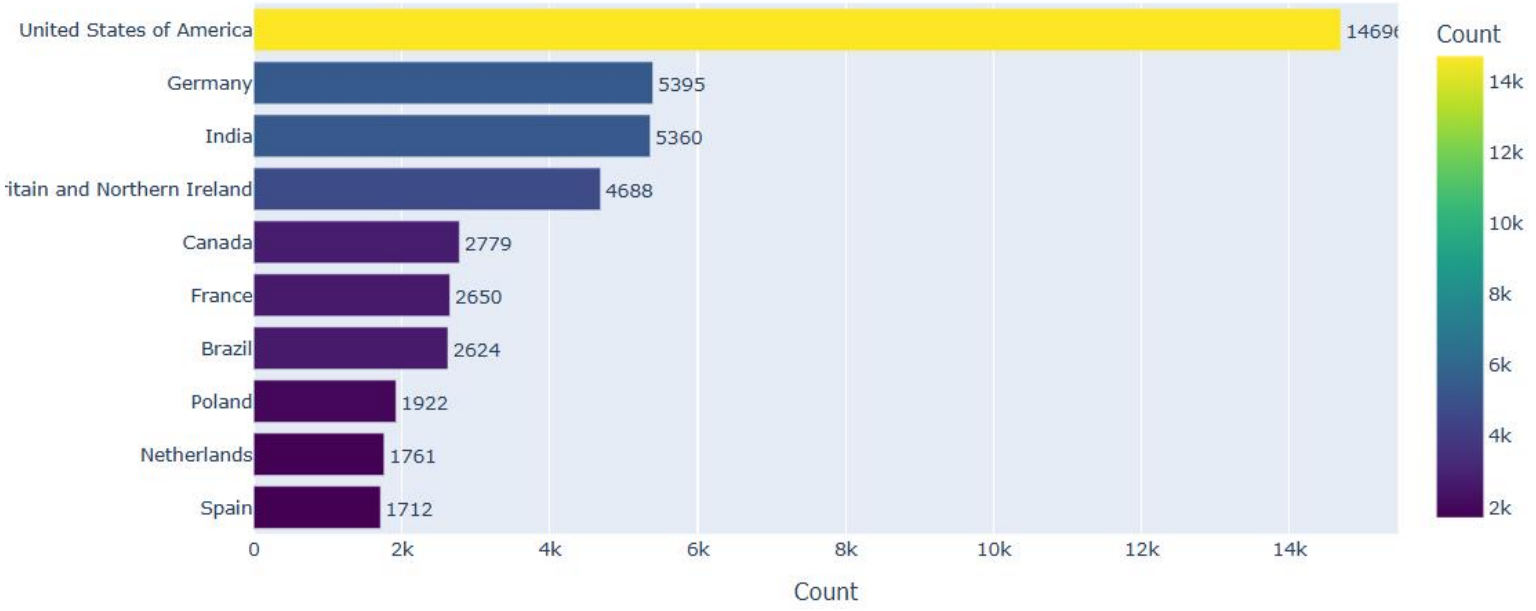
Folium



JOIN

```
pd.merge(stackoverflow_cleaned_df, country_status, how='left', on='Country')
```

Plotly



Country

Hypothesis Testing

Section	Feature(s) Tested	Method	Result
4.1	`Age` (>35 vs ≤35)	Permutation test	✅ Significant difference in hiring outcomes between groups
4.2	`Gender_Man` (Male vs Non-male)	Permutation test	✅ Gender is significantly associated with hiring outcome
4.3	`Education_Encoded`	Logistic regression	✅ Education level is significantly associated with hiring outcome
4.4	`PreviousSalary`	Logistic regression	❌ No significant association found
4.5	`MainBranch`	Chi-square test	✅ Professional developer status is significantly associated with hiring outcome
4.6	Individual feature contribution:	Residual-based permutation tests	
	`YearsCodePro`		❌ No significant additional predictive power
	`PreviousSalary`		❌ No significant additional predictive power
	`Education_encoded`		❌ No significant additional predictive power
	`ComputerSkills`		✅ Adds strong predictive power
4.7	Combined group: `YearsCodePro`, `PreviousSalary`, `Education_encoded`	Permutation test	✅ Jointly predictive when evaluated together

- Computer Skills stood out as the strongest individual predictor.

- Traditional indicators like salary and education showed limited individual predictive power but were jointly meaningful.
- Demographic variables such as age, gender, and developer status are significantly associated with employment outcomes.

Modeling

- BaseLine: Logistic Regression
- Improvement:
  - Logistic Regression with L1 Regulization
  - Random Forest
  - XGBoost
- We also used Neural Networks and KNN method to check the improvements with these models

Assumption

- We are fitting this model for the Computer Science job seekers.
  - First Criterion: Recall
  - Second Criterion: Accuracy
  - Extract the most important feature.

BaseLine: Logistic Regression

- Advantage:
  - Simplicity and Interpretability
  - Good Baseline for Comparison
- Limitation:
  - Limited to Linear Relationships
  - No Automatic Feature Selection
  - Prone to Overfitting

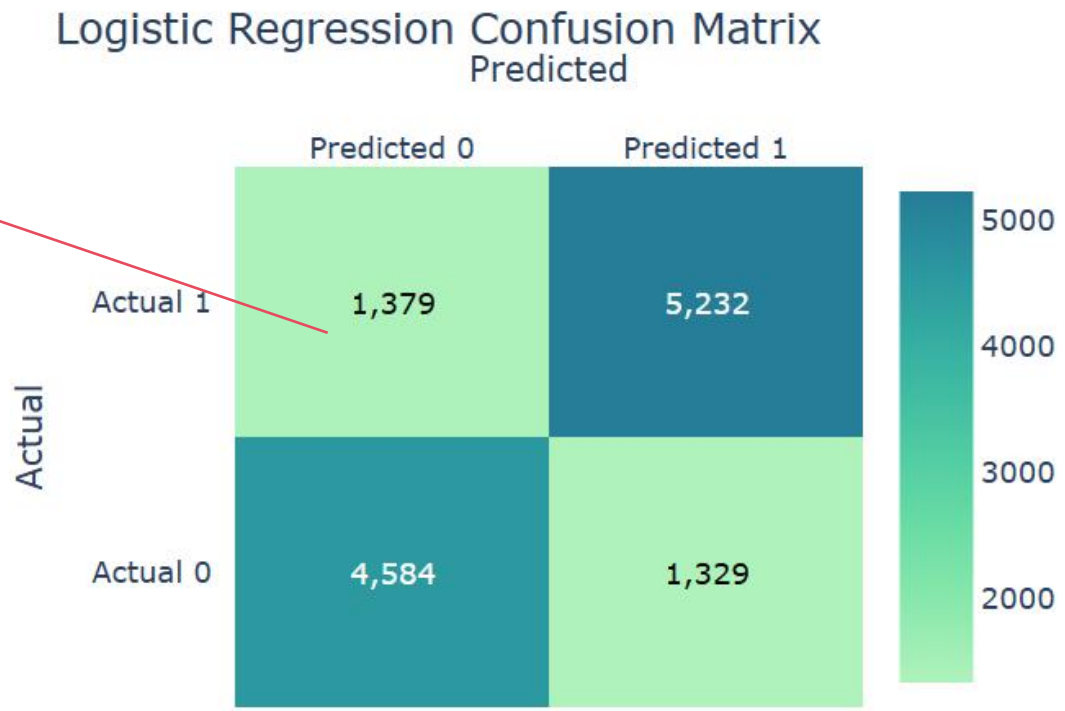
BaseLine: Logistic Regression

```
Accuracy : 0.7837751517087193
Train Accuracy : 0.7851681804571314
Precision: 0.7974394147233653
Train Precision: 0.7993205069476256
Recall    : 0.7914082589623355
Train Recall    : 0.7918242323400393
F1 Score : 0.7944123899180079
Train F1 Score : 0.7955547112462006
```

Criterions

Minimize this

No Overfitting



BaseLine: Logistic Regression

ComputerSkills	2.340963
MainBranch	-0.584933
Country_Development_Status	-0.240488
PreviousSalary	-0.197132
Employment	0.155238
Gender_Man	0.137852
Education_encoded	-0.122251
YearsCodePro	0.119376
Age	0.070770
Gender_NonBinary	0.048657
Gender_Woman	0.048140

dtype: float64

LR with L1 Regulization

- Advantage:
  - Automatic Feature Selection
- Limitation:
  - Still Limited to Modeling Linear Relationships

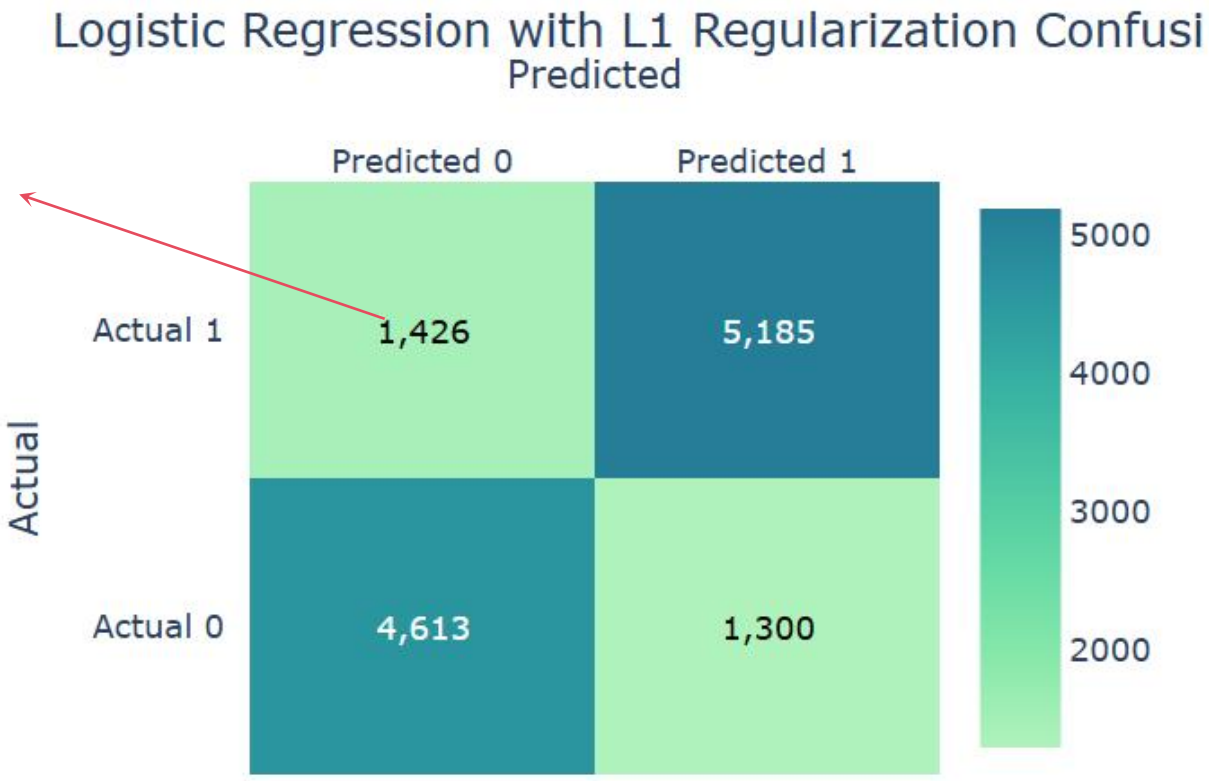
LR with L1 Regression

```
Accuracy : 0.7823379112104759
Train Accuracy : 0.7833915560435173
Precision: 0.7995373939861218
Train Precision: 0.8020377329252703
Recall    : 0.784298895779761
Train Recall    : 0.7828997126002117
F1 Score : 0.7918448381185095
Train F1 Score : 0.7923531775647281
```

Criterions

Minimize this

No Improvement



LR with L1 Regression

```
ComputerSkills       2.222012
MainBranch          -0.126133
Education_encoded   -0.099700
Employment           0.099034
YearsCodePro         0.081780
PreviousSalary      -0.079815
Gender_Man           0.050922
dtype: float64
```

Less Features are considered

Random Forest

- Advantage:
  - Captures Nonlinear Relationships
- Limitation:
  - Difficult in Tunning the Parameter

Grid Search

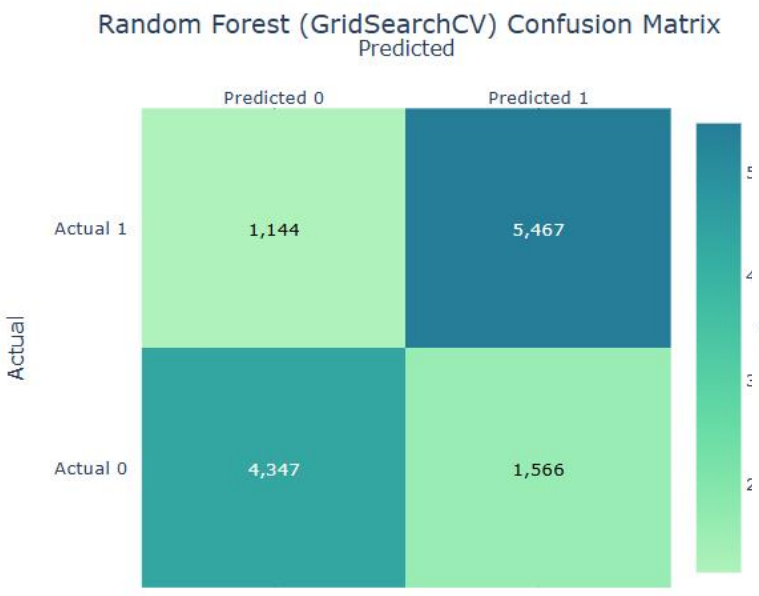
`n_estimators`: [100, 200, 300]
`max_depth`: [5, 10, 15]
`class_weight`: [None, 'balanced']

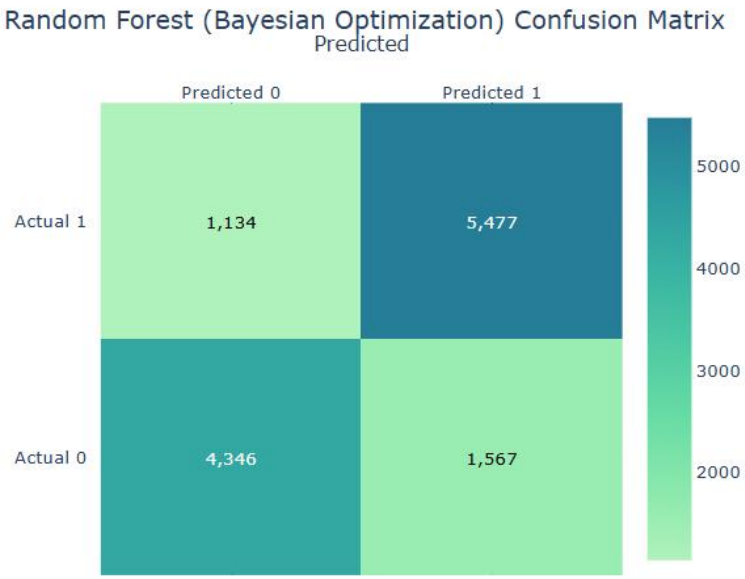
Best Param: {'class_weight': None,
'max_depth': 15,
'n_estimators': 100}

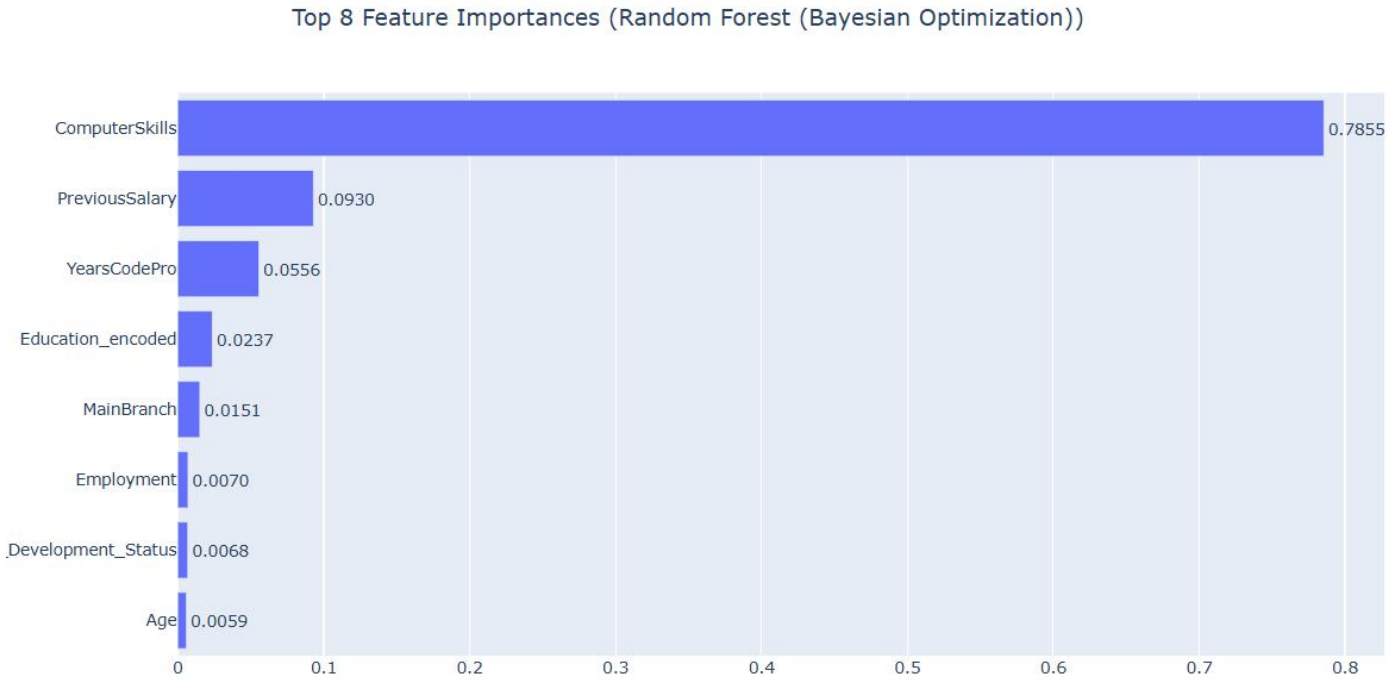
Bayesian Optimization

`n_estimators`: 80 to 200 (integer)
`max_depth`: 10 to 30 (integer)
`class_weight`: [None, 'balanced']

Best Param: {'class_weight': None,
'max_depth': 14,
'n_estimators': 140}







XGBoost

- Advantage:
  - Better Handling of Complex Feature Interactions
  - Faster Training with GPU Acceleration

XGBoost

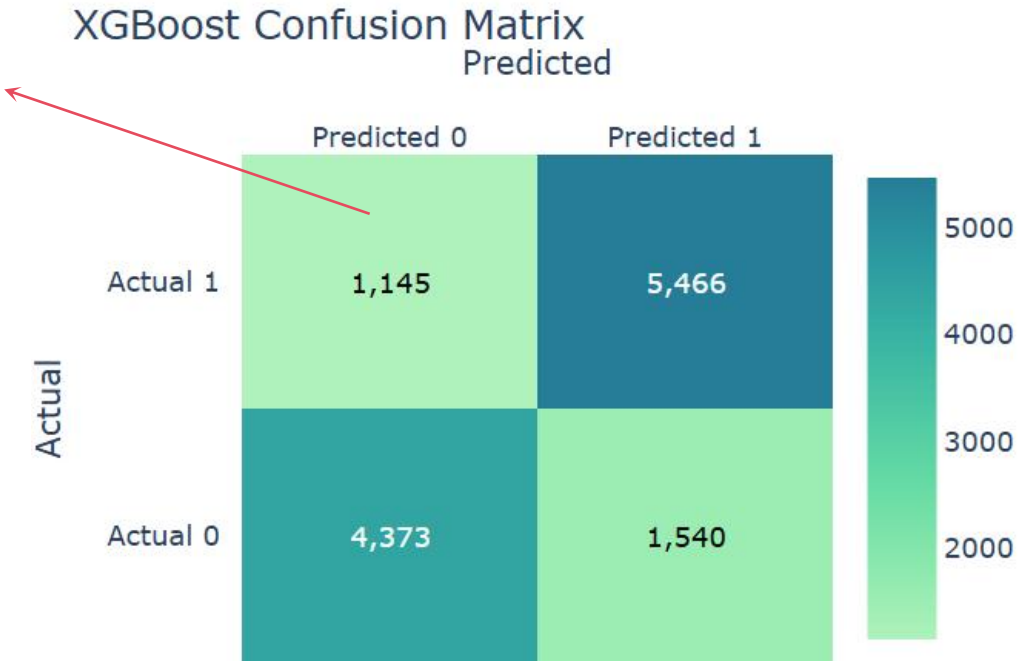
```
Accuracy : 0.7856116256786969
Train Accuracy : 0.7951891406327977
Precision: 0.780188409934342
Train Precision: 0.7896206156048676
Recall    : 0.8268038118287703
Train Recall    : 0.8342913326274392
F1 Score : 0.8028200044062569
Train F1 Score : 0.8113415710503089
```

Criterions

Minimize this

Little improvement in Accuracy

No risk in Overfitting



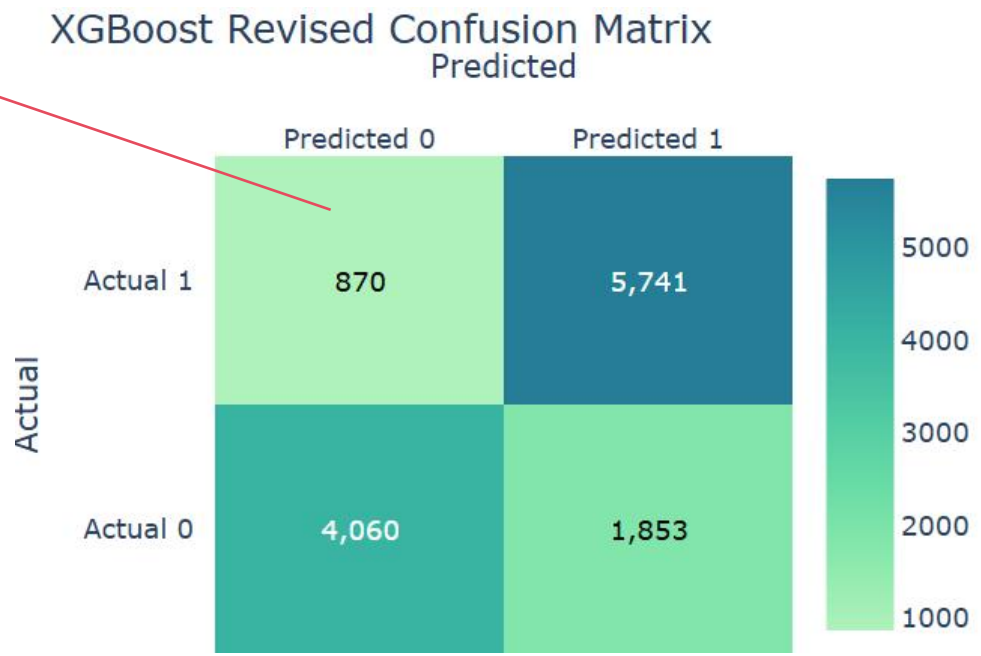
XGBoost with Tunning the Hyperparameters

```
Accuracy : 0.7825774512935164
Train Accuracy : 0.7982233755863859
Precision: 0.7559915722939162
Train Precision: 0.770284579748511
Recall    : 0.8684011495991529
Train Recall    : 0.8802752987445167
F1 Score : 0.8083069341781063
Train F1 Score : 0.8216151348298744
```

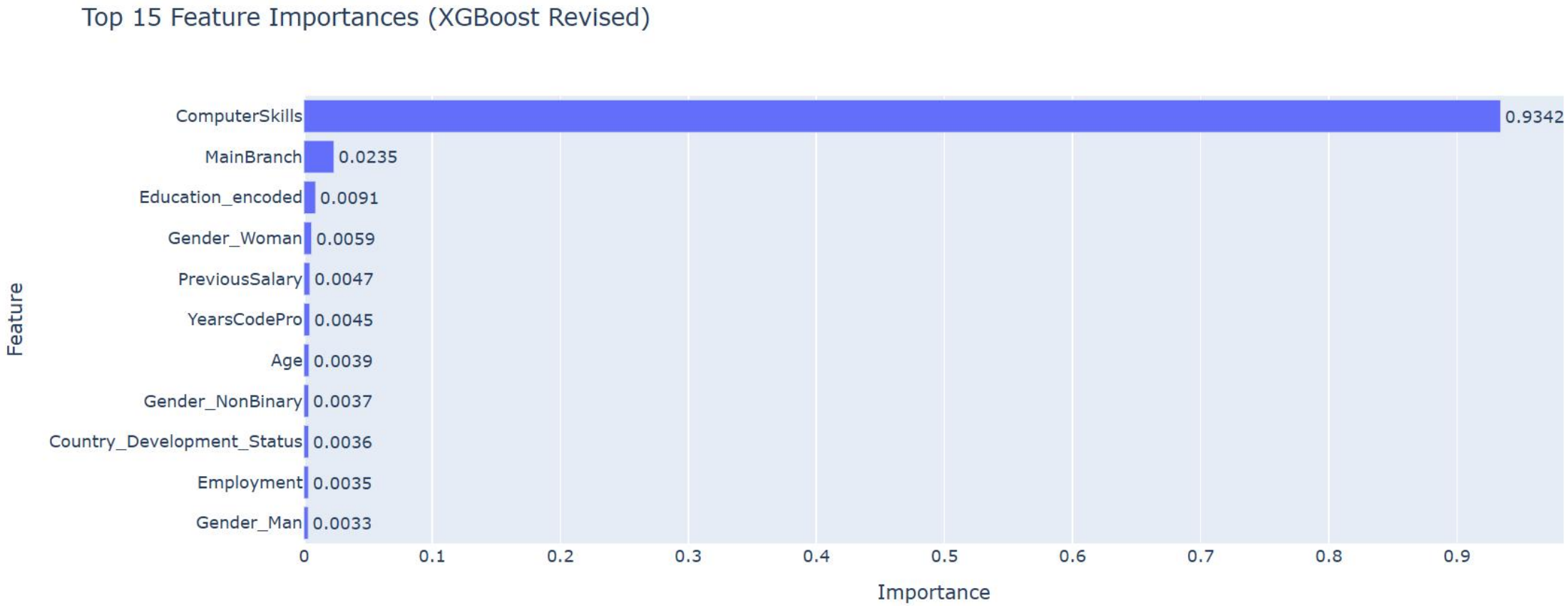
Criterions

Minimize this

Huge improvement in Recall



XGBoost with Tunning the Hyperparameters



Neural Networks

- Although Neural Networks is not suitable for Tabulate Data, but we decided to try on this method to see the accuracy and recall and compare it with other models. Also, we can extract the importance for features with “shap” method.

Neural Networks

	precision	recall	f1-score	support
0	0.79	0.74	0.77	5913
1	0.78	0.83	0.80	6611
accuracy			0.79	12524
macro avg	0.79	0.78	0.78	12524
weighted avg	0.79	0.79	0.79	12524

	precision	recall	f1-score	support
0	0.80	0.75	0.77	23651
1	0.78	0.83	0.81	26444
accuracy			0.79	50095
macro avg	0.79	0.79	0.79	50095
weighted avg	0.79	0.79	0.79	50095

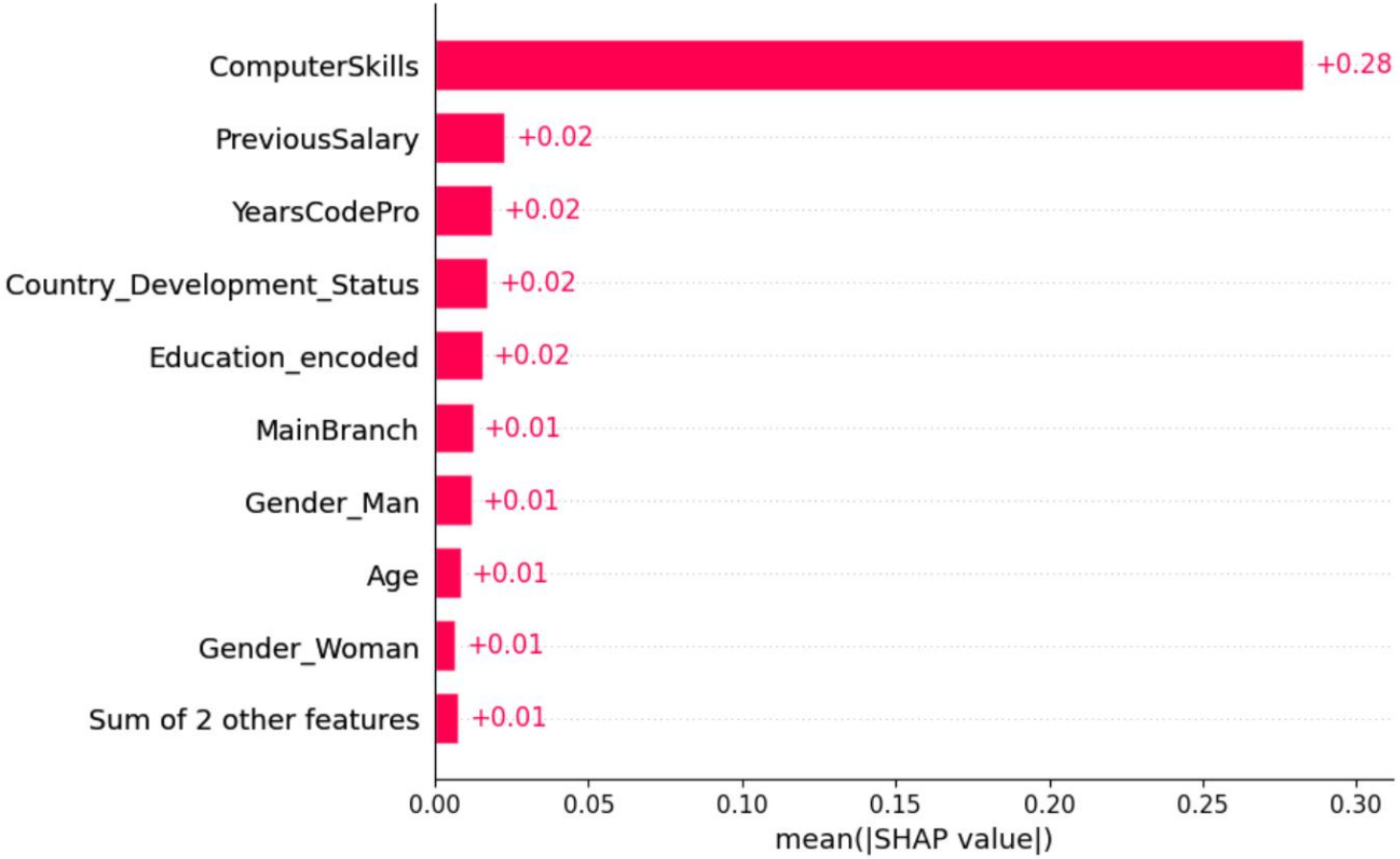
Criterions

Minimize this

```
[[4375 1538]
 [1140 5471]]
```

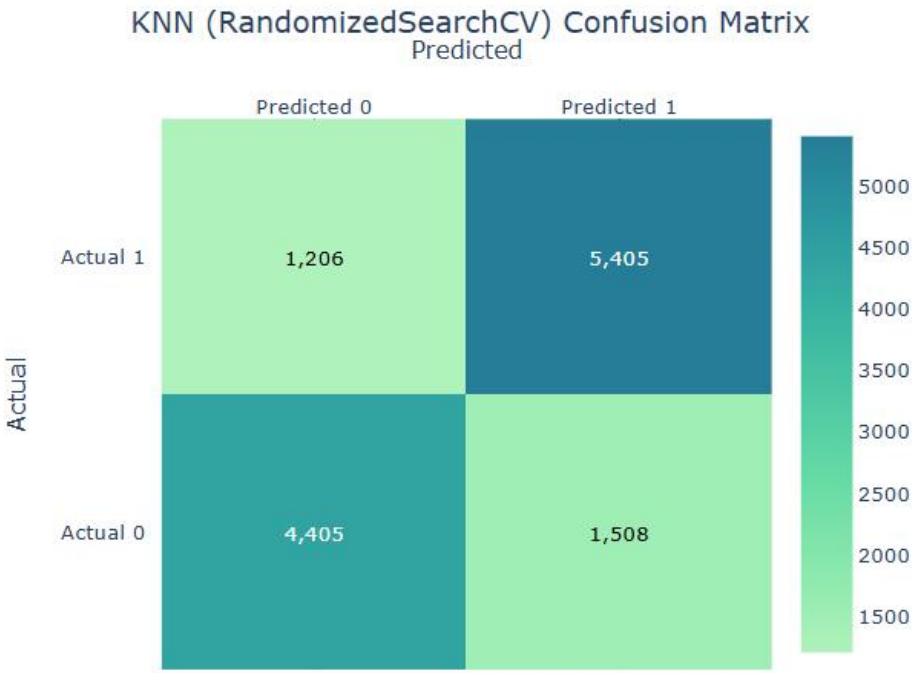
This result shows that NN doesn't improve much，
which indicates that NN is not suitable for Tabulate and Predictive Data.

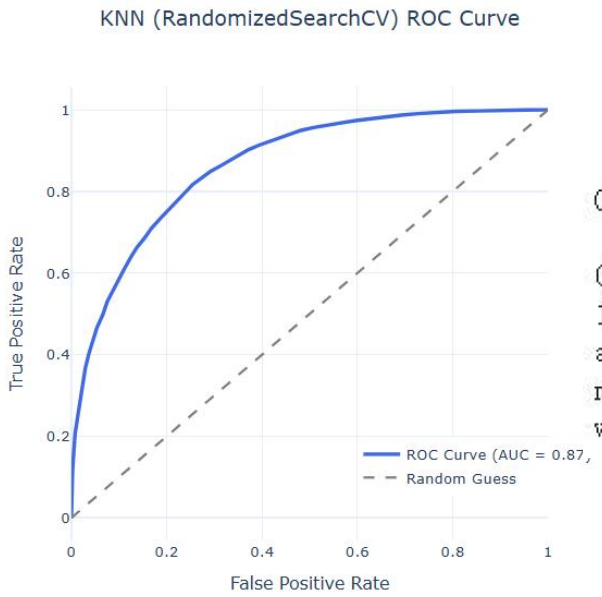
Neural Networks(shap)



KNN

- KNN is a distance-based, non-parametric algorithm that does not learn feature weights or model parameters.
- After hyperparameter tuning with RandomizedSearchCV, we achieved 78.33% accuracy and a strong AUC of 0.87.





Classification Report:

	precision	recall	f1-score	support
0	0.785065	0.744969	0.764491	5913.000000
1	0.781860	0.817577	0.799320	6611.000000
accuracy	0.783296	0.783296	0.783296	0.783296
macro avg	0.783463	0.781273	0.781906	12524.000000
weighted avg	0.783373	0.783296	0.782876	12524.000000

Implications/Insights

- Computer Skills consistently demonstrated strong predictive power in models that support feature importance analysis, which was also validated through hypothesis testing.
- Traditional features like education level had weak individual impact, but showed value when combined.
- Bayesian Optimization, guided by initial GridSearchCV, helped fine-tune models more efficiently.
- NN achieved the highest test accuracy (79%), even without tuning.
- Choose XGBoost model after tunning the hyperparameters, which kept a high accuracy and achieved a higher Recall score.

Challenges/Limitations

- Encoding the Country feature was challenging due to high cardinality; we simplified it by grouping countries by development status.
- Random Forest hyperparameter tuning was computationally expensive and time-consuming.
- During RandomizedSearchCV for KNN, the optimal number of neighbors was found to be 49—the upper limit of our search space. While this improved accuracy, we were hesitant to increase k further due to the risk of over-smoothing and potential underfitting, highlighting a trade-off in choosing k.

Potential Future Work

- Incorporate additional external datasets (e.g., company profiles, job market trends) to enrich the feature space and further improve model generalization.
- Apply NLP to textual fields.